

THE CONTACT OF TWO NOMINALLY FLAT ROUGH SURFACES

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Most models of surface contact consider the surface roughness to be on one of the contacting surfaces only. The authors give a general theory of contact between two rough plane surfaces. They show that the important results of the previous models are unaffected: in particular, the load and the area of contact remain almost proportional, independently of the detailed mechanical and geometrical properties of the asperities. Further, a single-rough-surface model can always be found which will predict the same laws as a given two-rough-surface model, although the required model may be unrealistic. It does not seem possible to deduce the asperity shape or deformation mode from the load-compliance curve.

INTRODUCTION

THE STUDY of friction, wear and thermal and electrical contact between solids has relied heavily on the idea that the real area of contact is extremely small. A newcomer to the subject might think it a truism that to understand these contact areas one must first understand the topography of the individual surfaces but, in practice, the classic texts by Holm (1)‡ and by Bowden and Tabor (2) use only the qualitative property that surfaces are rough. The reason is, of course, that, in the specific case of ideal plastic flow at the contact spots, one can calculate the total area of contact without a knowledge of the surface topography.

It is easy to show that contact between surfaces is rarely perfectly, or even predominantly, elastic. It does not, of course, follow that ideal plastic flow is the rule. If the asperities on the surfaces have rounded summits, then individual contacts will be analogous to contacts between a ball and a plane, as in a Brinell hardness test; but in this case (3) the load has to be increased about 200-fold to take the contact from the elastic limit to the fully plastic state. Even more important is the idea, introduced by Archard (4), that plastic flow involves permanent deformation and so cannot be repeated indefinitely; the contact geometry will gradually change until elastic contact becomes possible. In addition, when we wish to predict sliding temperatures (5), or thermal or electrical contact resistance (6) (7), a knowledge of the total area of contact

is inadequate; we need to know something of the number and sizes of individual contacts.

However, the surfaces of sliding solids are as complex as the surface of the earth and, indeed, geometrically rather similar to it. This appears to rule out any exact treatment of contact and instead we have to rely on models which, it is hoped, represent the essential features of the surface roughness. Fortunately, there are properties which are reasonably independent of the details of the model so that its fidelity is not of the first importance. Interest in models has been enhanced by the recent introduction of quantitative measurements of surface topography obtained by analysis of the output of a stylus profilometer. The analogue system used by Tallian *et al.* (8) has many advantages but digitizing the signal and recording it on paper-tape for subsequent analysis on a digital computer has proved more popular (9)–(11).

The one simplifying feature of surface roughness so far found is that, although the height of a particular asperity is random, the distribution of the asperity heights is rather close to Gaussian§ especially for ground or grit-blasted surfaces (8)–(13). Perhaps more important, the height distribution on a worn surface can be as shown in Fig. 1. This would usually be regarded as non-Gaussian but in contact the effect is Gaussian since the height of parts of the surface which do not touch is irrelevant. It is not yet known how common such behaviour is because systems for acquiring data of this sort readily are fairly recent, but other examples have already been found (13) (14).

If we can add to this characteristic distribution of

§ Or 'normal' but since this term has lost the meaning 'ideal' and acquired the meaning 'usual' we prefer to avoid it.

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‡ References are given in Appendix 4.

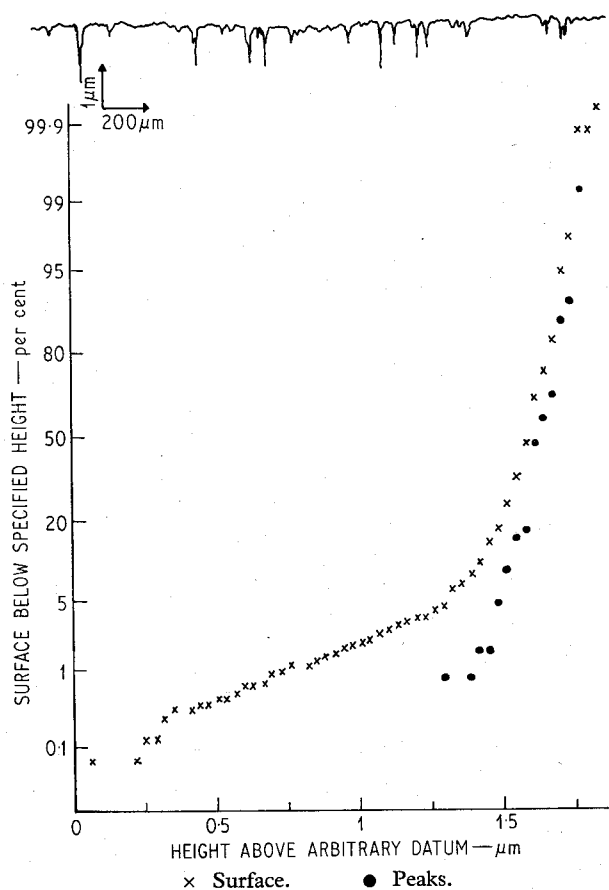


Fig. 1. Typical 'highly non-Gaussian' height distribution produced by wear. The non-Gaussian heights would not affect the contact properties, although the relevant standard deviation would change

asperity heights a characteristic asperity shape, we would have a model surface which we could use to predict some properties of real surfaces. Archard (10) has recently suggested that surface features of different scales may be essential in a realistic model. However, the object of this paper is to attempt to settle a particular controversy which seems unlikely to depend on such questions, i.e. whether the contact problem can be simplified by assuming only one of the contacting surfaces to be rough or whether treating both surfaces as rough introduces new features.

While the only other work in which both surfaces were rough (15) gave results for one specific case, we shall treat several cases and present some general conclusions. It must be emphasized that this is a purely theoretical paper; the analysis is based on measured properties of surfaces but the predictions of the theory have not been verified experimentally.

GENERAL THEORY

A rough surface is represented by an array of identical asperities differing only in their heights above a reference plane, which is conveniently taken to pass through the mean of the height distribution. We shall consider the

contact between two such surfaces. The pairs of asperities will not be aligned so that the usual contact will be between the shoulders of the two hills.

Let the distribution of asperity heights be $\phi(z)$, and let the shape of each asperity with respect to its summit be $y = f(\rho)$, where ρ and y are the horizontal and vertical distances from the summit. Consider a particular asperity on the first surface, with height z_1 and all the asperities of height z_2 on the second surface; a typical one will be a horizontal distance r away (see Fig. 2). Since the asperities are identical, any contact must lie symmetrically between their centres and, if they are convex, the contact will be centred a distance $r/2$ from the individual centres, at which point the heights above the respective mean planes will be

$$z_1 - f(r/2); \quad z_2 - f(r/2)$$

For contact, the separation d of the mean planes must be less than the sum of these heights, i.e.

$$d < z_1 + z_2 - 2f(r/2)$$

and we may define the 'interference' at the contact as

$$w = z_1 + z_2 - 2f(r/2) - d$$

During loading, the area of contact A and the vertical force P will depend only on the interference w and the misalignment r , i.e.

$$A = A(w, r); \quad P = P(w, r)$$

(During unloading they may involve also the maximum interference achieved: we shall not consider this case.)

The number of asperities with heights in the range z_2 to $z_2 + dz_2$ situated between r and $r + dr$ from the first asperity will be

$$\eta \cdot 2\pi r dr \phi(z_2) dz_2$$

where η is the surface density of asperity peaks on either surface, and so the expected value of the force on the first asperity due to the second surface will be

$$\int_{z_2} \int_r 2\pi \eta P(w, r) \phi(z_2) r dr dz_2$$

where the r integration will, in principle, be limited by the extent of the surfaces but in practice may be taken from 0 to ∞ . (While all the asperities on the second surface are considered in the integral, very few make any contribution since w will be negative and $P(w, r)$ will vanish, but see Appendix 1.)

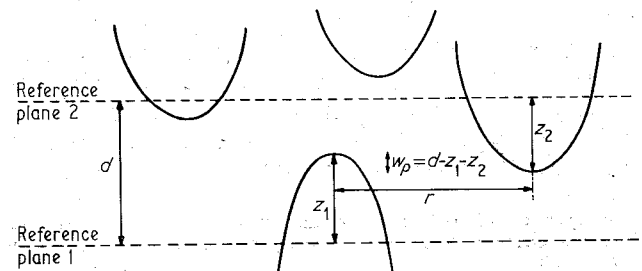


Fig. 2. Contact of two rough surfaces. Since all asperities are assumed to have the same shape, contact will first occur midway between their centres

If the apparent area of contact is $\mathcal{A}$ there will be $\mathcal{A}\eta\phi(z_1)dz_1$ asperities on the first surface with heights between z_1 and z_1+dz_1 . Thus, the expected total force between the surfaces will be

$$\bar{P}(d) = 2\pi\eta^2\mathcal{A} \int_{z_1} \int_{z_2} \int_r P(w, r)\phi(z_1)\phi(z_2)r dr dz_1 dz_2$$

The force between a pair of asperities with given heights does not depend on the individual heights z_1 and z_2 but only on their sum (z_1+z_2), and therefore only the statistical distribution of the sum of pairs of heights can affect $\bar{P}(d)$. The distribution of the sum can be found from the individual distributions $\phi(z_1)$ and $\phi(z_2)$ by the standard processes of mathematical statistics. This may be simple, or not, but for our purposes it is convenient to ignore this step and start from the distribution of the sums of heights as the basic distribution. Thus, we set $z = z_1+z_2$, and assume it has a distribution $\phi_0(z)$. The expected total force is then

$$\bar{P}(d) = 2\pi\eta^2\mathcal{A} \int_z \int_r P(w, r)\phi_0(z)r dr dz \quad (1)$$

Now, $w = z-d-2f(r/2) = w_p-2f(r/2)$ where $w_p = z_1+z_2-d$ is the overlap between the peaks of the pair of asperities considered: because of misalignment the peaks may overlap without there being any contact. If we define the result of the r -integration as

$$P_0(w_p) = 2\pi\eta \int_0^\infty P(w_p-2f(r/2), r)r dr \quad (2)$$

then

$$P(d) = \eta\mathcal{A} \int_d^\infty P_0(z-d)\phi_0(z) dz \quad (3)$$

In the same way, if we write

$$A_0(w_p) = 2\pi\eta \int_0^\infty A(w_p-2f(r/2), r)r dr \quad (4)$$

the expected total area of contact is

$$\bar{A}(d) = \eta\mathcal{A} \int_d^\infty A_0(z-d)\phi_0(z) dz \quad (5)$$

If we analyse the contact between a rough surface and a plane in a similar way, taking the laws describing the deformation of a single contact to be $P = P_0(w)$, $A = A_0(w)$, we arrive at equations (3) and (5) precisely. The difference, then, between a single rough surface and both surfaces being rough is simply (a) the replacement of the actual single-contact laws by generalized laws, found by integrating the actual deformation laws over all possible radial misalignments and (b) the replacement of the distribution of asperity heights by the distribution of sums of pairs of heights. Therefore, there is nothing essentially different about models in which both surfaces are rough.

However, the behaviour may sometimes be more readily explained on one hypothesis than on the other. For example, the contact between a spherical asperity and a plane gives $A \propto w$; between a cone and a plane $A \propto w^2$. Higher powers seem unlikely. The effect of radial averaging (see below) is to lead to $A \propto w_p^2$, $A \propto w_p^4$ respectively, so that a single rough surface covered with conical asperities behaves like a pair of rough surfaces covered with spheres, but a pair

of rough surfaces covered with cones has no plausible equivalent (although suitable cusp-like asperities could be invented).

SPECIFIC MODELS

We assume first that the surfaces are covered with paraboloidal asperities (which in the usual Hertzian approximation includes spheres and radially symmetrical ellipsoids as well). The asperity shape is then $y = f(\rho) = \rho^2/2\beta$ where β is the radius of curvature at the peak. Thus $2f(r/2) = r^2/4\beta$. We consider two modes of deformation which we shall call 'elastic' and 'plastic': there seems no justification for the mathematical complexities which more exact solutions would involve.

Elastic deformation

The Hertzian solution for the contact of two paraboloids applies here only approximately because of the misalignment, which means that the normal force is not acting vertically and there must also be a tangential component. However, the slopes on real surfaces are so small that the errors will be small and we shall ignore them. We assume the asperity deformation laws to be exactly

$$A = \pi \left(\frac{\beta}{2}\right) w; \quad P = \left(\frac{4}{3}\right) E' \left(\frac{\beta}{2}\right)^{1/2} w^{3/2}$$

where the term 2β is the sum of the asperity curvatures and E' is the usual composite elastic modulus, given by

$$\frac{1}{E'} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$$

Then from equation (2) the average force-displacement law is

$$P_0(w_p) = \left(\frac{8}{3}\right) \pi\eta E' \left(\frac{\beta}{2}\right)^{1/2} \int_0^\infty \left\langle w_p - \frac{r^2}{4\beta} \right\rangle^{3/2} r dr$$

where $\langle \rangle$, a 'Macauley bracket' (16), denotes that the term is set to zero if the quantity enclosed is negative. Thus

$$P_0(w_p) = (16\sqrt{2}/15) \pi\eta E' \beta^{3/2} \langle w_p \rangle^{5/2} \quad (6)$$

Then the total load, using equation (3), is

$$\bar{P}(d) = (16\sqrt{2}/15) \pi\eta^2 E' \beta^{3/2} \mathcal{A} \int_d^\infty (z-d)^{5/2} \phi_0(z) dz$$

and putting $z = s\sigma$ we get

$$\bar{P}(d) = (16\sqrt{2}/15) \pi(\eta\beta\sigma)^2 E' \sqrt{\left(\frac{\sigma}{\beta}\right)} \mathcal{A} F_{5/2} \left(\frac{d}{\sigma}\right) \quad (7)$$

where $F_n(u) = \int_u^\infty (s-u)^n \phi^*(s) ds$

and $\phi^*(s)$ is the standardized height distribution so that $\phi^*(s) ds = \phi(z) dz$.

The expression for $\bar{P}(d)$ may, in fact, be obtained more neatly without introducing $P_0(w_p)$ by changing the order of integration in equation (1); but the parallel with single-rough-surface models is then lost.

Similarly, we find the area $\tilde{A}$ is given by

$$\tilde{A}(d) = \pi^2(\eta\beta\sigma)^2 \mathcal{A} F_2\left(\frac{d}{\sigma}\right) \quad (8)$$

while the expected number of contacts is

$$\tilde{N}(d) = 4\pi(\eta\beta\sigma)\eta \mathcal{A} F_1\left(\frac{d}{\sigma}\right) \quad (9)$$

so we have expressed P , A and N in terms of integrals of the distribution of asperity heights. We can compare these results in several ways with those for contact of a single rough surface with a plane. In the first place, we can consider an 'equivalent' single rough surface, defined as one whose asperity curvature is the sum of the two curvatures and whose height distribution is that of the sum of the heights of the original pair. In fact, the behaviour of the 'equivalent' surface is exactly that of the original pair *with their asperities aligned* ('non-genuine' two-rough-surface models of this type were discussed in (17)), so this comparison is a study of the effect of permitting misalignment. We can also compare the behaviour with that of one of the original rough surfaces touching a plane.

We divide the force equations by the nominal area $\mathcal{A}$ to get the nominal pressure $p = \tilde{P}/\mathcal{A}$. For the first case, by substituting $P_0(w) = (4/3)E'\beta_e^{1/2}w^{3/2}$ in equation (3), we get

$$p_e(d) = \left(\frac{4}{3}\right)(\eta\beta_e\sigma)E' \sqrt{\left(\frac{\sigma}{\beta_e}\right)} F_{3/2}\left(\frac{d}{\sigma}\right) = K F_{3/2}\left(\frac{d}{\sigma}\right) \quad (10)$$

$$\text{where } \beta_e = \frac{\beta}{2} \text{ and } K = \frac{2\sqrt{2}}{3} \cdot (\eta\beta\sigma) \cdot E' \sqrt{\left(\frac{\sigma}{\beta}\right)}$$

while when both surfaces are rough we obtain from equation (6):

$$p(d) = \frac{8\pi}{5} \cdot (\eta\beta\sigma) \cdot K \cdot F_{5/2}\left(\frac{d}{\sigma}\right) \quad (11)$$

At this point a specific distribution of asperity heights must be chosen. We shall use the Gaussian distribution because experimental work suggests that actual surface distributions approximate well to it. A short table of the functions $F_n(h)$ for this case is given in Appendix 2. By elementary statistics, if the height distribution on each surface is Gaussian with standard deviation σ_1 , the distribution of sums of heights is also Gaussian, with standard deviation $\sigma = \sigma_1\sqrt{2}$. Therefore, considering one of the same surfaces touching a plane, we find, as in equation (10),

$$\left. \begin{aligned} p_1(d) &= \left(\frac{4}{3}\right)(\eta\beta\sigma_1)E' \sqrt{\left(\frac{\sigma_1}{\beta_1}\right)} F_{3/2}\left(\frac{d}{\sigma_1}\right) \\ &= 2^{-1/4} K \cdot F_{3/2}\left(\frac{d\sqrt{2}}{\sigma}\right) \end{aligned} \right\} \quad (12)$$

To compare these three cases we need to choose a value for $\eta\beta\sigma = \eta\beta\sigma_1\sqrt{2}$. Experiments with the equipment described by Greenwood and Williamson (9) suggest that $\eta\beta\sigma_1$ is reasonably constant with a value of 0.03–0.05, so we shall assume $\eta\beta\sigma = 0.05$. The value of K is less important, but to get definite scales we select the plausible

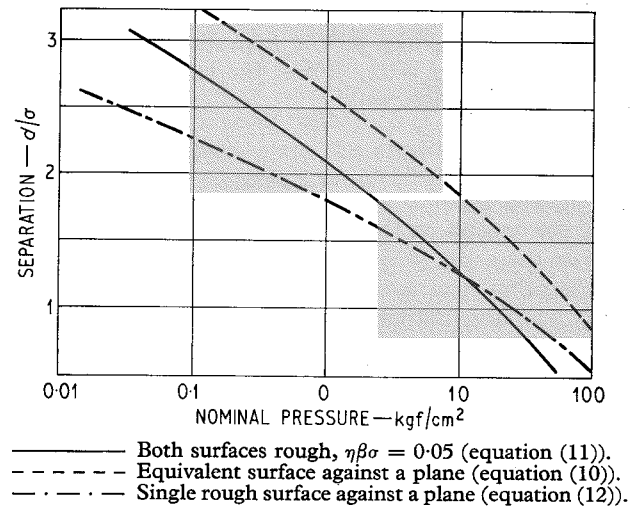


Fig. 3. Relation between load and separation

value 1000 kgf/cm². The results are shown in Fig. 3. The curve for two rough surfaces against a plane differs from that for the equivalent surface against a plane only in that the separation is reduced—a not unexpected effect of permitting misalignment of the asperities. The slopes of the log(load)–separation curves, which are all that can be detected experimentally, remain unchanged. On the other hand when one of the rough surfaces is loaded against a plane, the slope is much less—a much more significant change.

If we make similar calculations for the expected area of contact as a function of separation, we can then plot the area as a function of load (Fig. 4). The points for a single rough surface and for two rough surfaces fall on the same, almost straight, line. Thus, again, we obtain the basic statistical law of contact, that the area is proportional to the load: and we can think of an 'elastic hardness' as in (10)

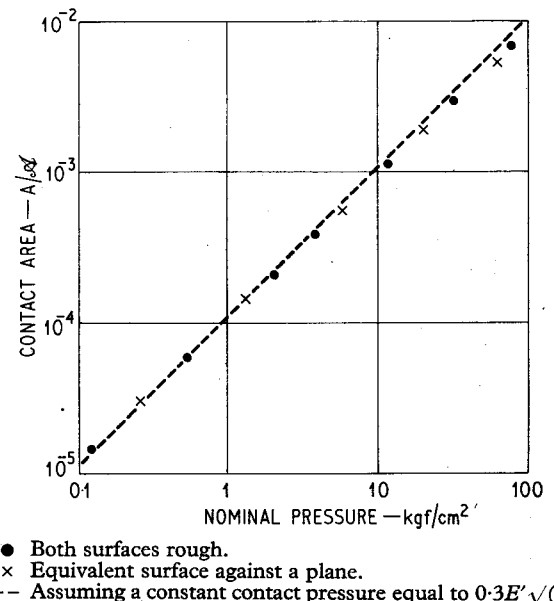


Fig. 4. Relation between area of contact and load

equal to $0.3E'\sqrt{(\sigma/\beta_e)}$ in terms of the equivalent properties for the pair of surfaces. In terms of the parameters for a single surface, the 'elastic hardness' becomes approximately $0.5E'\sqrt{(\sigma_1/\beta)}$.

Plastic deformation

It may be shown that the intersection of two equal paraboloids

$$y_1 = d - z_1 + \frac{\rho_1^2}{2\beta}, \quad y_2 = z_2 - \frac{\rho_2^2}{2\beta}$$

is an ellipse whose horizontal projection is a circle of radius

$$\left[\beta(z_1 + z_2 - d) - \left(\frac{r}{2}\right)^2 \right]^{1/2}$$

where r is the distance between the paraboloid axes.

The interference at the centre of the circle is $w = z_1 + z_2 - d - (r/2)^2/2\beta$, so the radius is just $(\beta w)^{1/2}$ and the area $\pi\beta w$. While the actual force will be inclined to the vertical, it seems reasonable to ignore the horizontal component and take the vertical component to be proportional to the horizontal projection of the contact area, i.e.

$$P = \pi\beta H < w >$$

where H is a mean pressure related to the hardness. The integration is similar to that for elastic deformation, and we find

$$\bar{P}(d) = 2\pi^2(\eta\beta\sigma)^2 H \mathcal{A} F_2 \left(\frac{d}{\sigma} \right) \quad (13)$$

The result for aligned asperities is

$$\bar{P}(d) = \pi(\eta\beta\sigma) H \mathcal{A} F_1 \left(\frac{d}{\sigma} \right) \quad (14)$$

Load-separation curves according to equations (13) and (14) are shown in Fig. 5. They are very similar to those already given for elastic contact.

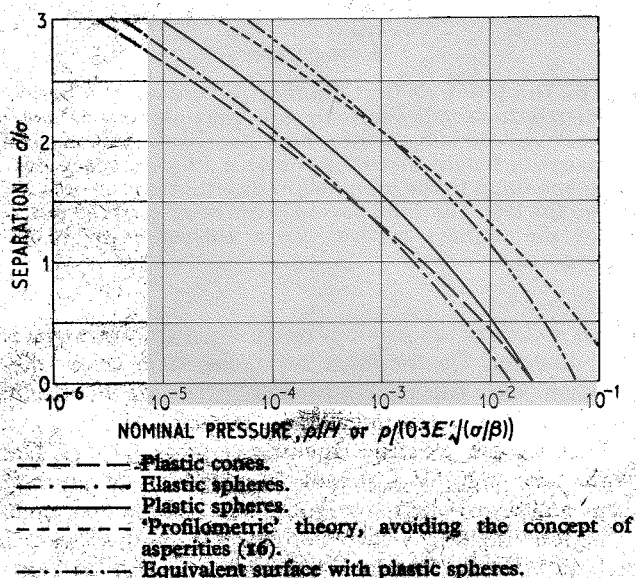


Fig. 5. Independence of load-separation relation and deformation model

Conical asperities

Now consider conical asperities whose sides have a slope α . It can be shown that geometrically a pair of these, with axes separated by r and a peak overlap w_p , will intersect in an ellipse whose horizontal projection has semi-axes

$$\frac{1}{2}w_p \cot \alpha, \quad \frac{1}{2}\sqrt{(w_p^2 \cot^2 \alpha - r^2)}$$

provided $w_p > r \cot \alpha$. The physical deformation law is unknown, so we shall take the area of contact to be this projected ellipse, and the vertical force to be the area multiplied by an effective hardness H : i.e.

$$P = \frac{\pi H}{4} \cdot w_p \cot \alpha \cdot (w_p^2 \cot^2 \alpha - r^2)^{1/2}$$

This may be written in terms of the interference $w = w_p \cot \alpha - r$, but there is no advantage in doing so. Carrying out the radial averaging process, we find

$$\begin{aligned} P_0(w_p) &= 2\pi\eta \cdot \int_0^{w_p \cot \alpha} \frac{\pi H}{4} \cdot w_p \cot \alpha \cdot (w_p^2 \cot^2 \alpha - r^2)^{1/2} r dr \\ &= \frac{1}{2}\pi^2 \eta H w_p^4 \cot^4 \alpha \cdot \int_0^{\pi/2} \cos \theta \cdot \sin \theta \cdot \cos \theta d\theta \\ &= \frac{1}{8}\pi^2 \eta H \cot^4 \alpha \cdot w_p^4 \end{aligned}$$

Thus, again assuming a Gaussian height distribution, we find

$$\begin{aligned} \bar{P}(d) &= \eta \mathcal{A} \int_d^\infty \frac{1}{6} \pi^2 \eta H \cot^4 \alpha \cdot (z-d)^4 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2/2\sigma^2} dz \\ &= \frac{1}{6} \pi^2 \eta^2 \mathcal{A} H \cot^4 \alpha \cdot \sigma^4 F_4 \left(\frac{d}{\sigma} \right) \quad (15) \end{aligned}$$

In order to compare this result with the previous ones, we need to harmonize the two models of surface roughness. We might argue that a paraboloidal asperity of height h and radius of curvature β is 'equivalent' to a cone inscribed in it (M. El Refaie, private communication) which leads to $\tan \alpha = \sqrt{(h/2\beta)} \sim \sqrt{(\sigma/2\beta)}$. Assuming this, equation (14) becomes

$$\bar{P}(d) = \frac{2}{3} \cdot \pi^2 \mathcal{A} H \cdot (\eta\beta\sigma)^2 F_4 \left(\frac{d}{\sigma} \right)$$

which is compared with equation (11) in Fig. 5.

DISCUSSION

It appears from Fig. 5 that, as long as the height distribution is Gaussian, the mode of deformation, asperity shape, and whether the asperities are on one or both surfaces are all unimportant. In all cases the area of real contact is almost proportional to the load and the load-compliance curve is almost straight on a log (load)-linear compliance plot, with a slope such that a movement of one standard deviation of the joint height distribution gives a load increase by a factor of about 50. This corresponds very roughly with a movement equal to the c.l.a. of each surface. The analysis requires a single type of asperity on both surfaces and, to this extent, it is unrealistic. Examination of the theory shows that the height distribution on the two surfaces need not be the same; only the joint

height distribution is important. However, it seems unlikely that a pair of surfaces with similar asperity shapes but different height distributions would occur in practice.

Of course height distributions other than the Gaussian often occur and the general theory can immediately be applied to them. For example, Zhuravlev (18), in possibly the earliest proof that the real area of contact could be almost proportional to the load even though individual contacts behaved elastically, used a model of aligned, spherical asperities, with a linear height distribution

$$\phi(z) = \frac{2}{L^2}(L-z), \quad 0 < z < L$$

The distribution of sums of heights may be shown to be

$$\phi_0(z) = \frac{2}{3L^4}(2L-z)^3 \quad \text{for } L < z < 2L$$

and a more complicated expression for $0 < z < L$ (Fig. 6). The radially averaged law for the contact of elastic spheres has already been found (equation (6)) and, combining the two, we obtain the generalization of Zhuravlev's result for non-aligned asperities:

$$\bar{P}(d) = \frac{16\sqrt{2}}{15} \pi \eta^2 E' \beta^{3/2} \int_d^{2L} (z-d)^{5/2} \cdot \frac{2}{3L^4} (2L-z)^3 dz \quad \text{for } d > L$$

$$\text{i.e.} \quad \bar{P}(d) \propto (2L-d)^{13/2}$$

Similarly $\bar{A}(d) \propto (2L-d)^6$ and so $\bar{A} \propto \bar{P}^{12/13}$.

For aligned asperities we omit the radial averaging and obtain

$$\bar{P}(d) = \frac{2\sqrt{2}}{3} \eta E' \beta^{1/2} \int_d^{2L} (z-d)^{3/2} \cdot \frac{2}{3L^4} (2L-z)^3 dz \quad (d > L)$$

$$\text{i.e.} \quad \bar{P}(d) \propto (2L-d)^{11/2}$$

and similarly $\bar{A}(d) \propto (2L-d)^5$; $\bar{A} \propto \bar{P}^{10/11}$

Thus the effect of misalignment is small, but leads to a slight improvement in the proportionality compared to Zhuravlev's result.

Ling (15) used the simple rectangular height distribution

$$\phi(z) = \frac{1}{2L}, \quad -L < z < L$$

The distribution of sums of heights is

$$\phi_0(z) = \frac{1}{4L^2}(2L-|z|), \quad -2L < z < 2L$$

(see Fig. 6b). If we combine this with a model of conical asperities deforming as in the previous section, we get

$$\bar{P}(d) = \frac{1}{6} \pi^2 \eta^2 \mathcal{A} H \cot^4 \alpha \int_d^{2L} (z-d)^4 (2L-z) dz \quad (z > L)$$

i.e.

$$\bar{P}(d) = \frac{1}{180} \pi^2 \eta^2 \mathcal{A} H \cot^4 \alpha \cdot (2L-d)^6 \quad (16)$$

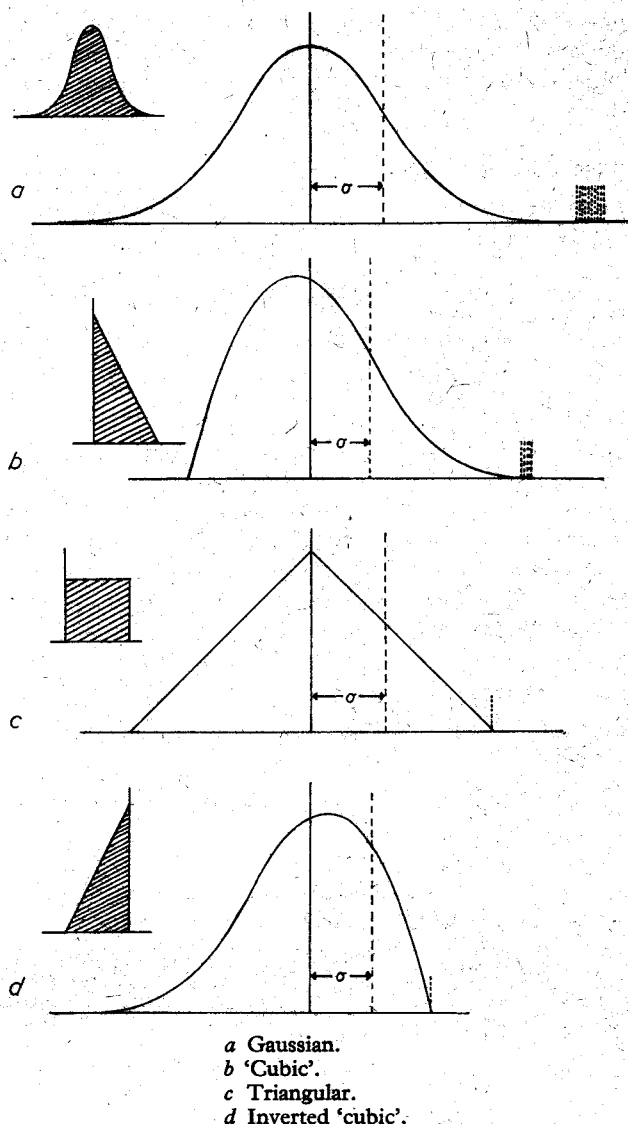


Fig. 6. The point of first contact: whenever the height distribution falls gently to zero, the point of first contact will be subject to a high degree of scatter. The asperity height distributions on a single surface corresponding to the joint height distributions are shown on the left. The zone dotted is the inter-quartile range of the highest asperity on a surface with 10 000 asperities

which is close to the previous result, although the model is very different. On the other hand, the same height distribution with spherical asperities deforming plastically gives $\bar{P} \propto (2L-d)^4$, a much lower dependence.

Thus we can construct almost any power law we like (with index > 1) by a suitable combination of asperity shape and distribution law. The actual deformation mode is the least important part. For example, Ling suggested that 45° conical asperities undergo a process of compression followed by the tip shearing off and the force building up again from zero. This still leads to a radially

averaged law $P_0 \propto w_p^4$ and, in fact, the constant of proportionality is rather close to the result found above for simple compression. Thus, the final result is indistinguishable from equation (16) with $\alpha = 45^\circ$.

Actual measurements of asperity height distributions do not support either of the above forms which, indeed, were postulated long before adequate surface data were available. They are important, however, in indicating what may be expected from a non-Gaussian distribution. In particular, approximate proportionality can be obtained between contact area and load, provided the number of asperities at a given height does not increase with height (Fig. 6c). The effect of both surfaces being rough is to increase tolerance of non-Gaussian distributions.

The load-compliance curve shows more variations, but it is impossible to work back from a given load-compliance curve and deduce a mode of deformation. In practice, it may also be difficult to determine experimentally exactly what the load-compliance relation is. Ling (15) found that his experimental results were of the form $P \sim (L-d)^n$ with $n = 7-8$. But when a typical set of his results is replotted on log (load)-linear compliance scales (Fig. 7), they could be interpreted as resulting from a Gaussian distribution. The scatter of the experimental points might well mask a slight curvature: the slope of the curve implies a standard deviation of the joint height distribution equal to $10 \mu\text{in}$, in good agreement with the quoted c.l.a. of $7 \mu\text{in}$ for each of the surfaces involved.

The point of first contact

We have found two types of load-compliance law: a power law form in which the point of first contact is taken as origin, and an approximately logarithmic form in which the point of first contact does not appear and, indeed, does

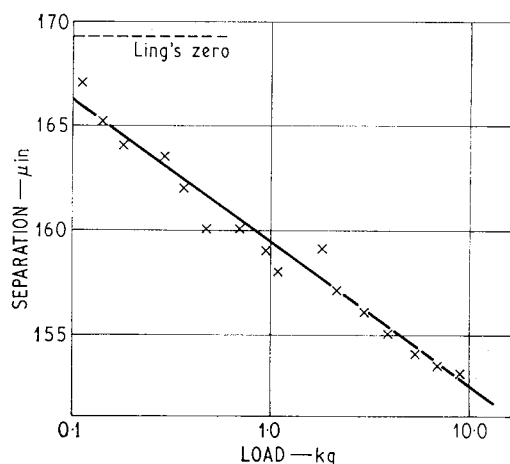


Fig. 7. Replot of Ling's experimental results (Fig. 9 of (15)) Ling's original criticism of single-rough-surface models was based on the discrepancy between experimental and theoretical curves on a log-log plot. On a log-linear plot, the discrepancy does not appear and it seems that it arises from an invalid extrapolation to zero load to find the origin for the log-log plot

not exist: small but non-zero loads are predicted whatever the separation. The practical experimenter may dismiss this but it indicates a very practical problem. A more general question is whether there is a finite possibility of an asperity 0.01 in high on a ground surface of $100 \mu\text{in}$ c.l.a., or whether the possibility is exactly zero. (For a Gaussian distribution the probability would be 10^{-1000} , so the distinction is a fine one.) Some of the argument arises from a confusion between a sample and a population: the highest asperity on an actual surface will certainly have a finite height but so will the largest of a sample of 10 000 asperities from a Gaussian population. In fact, the expected greatest height will be about 4σ above the mean, the same as the maximum possible height for the cubic distribution of Fig. 6b. A point often overlooked, however, is that the highest member of a sample from a *finite* distribution, such as the cubic distribution, is also subject to sampling variations. In fact, we find that the inter-quartile range (19) for the highest of a sample of 10 000 from the cubic distribution—the range in which the probability of finding the highest is one half—is from 3.49σ to 3.66σ (see Appendix 3). The corresponding inter-quartile range for the Gaussian is 3.68σ to 4.04σ . The only important difference is the extent of the spread; twice as great for the Gaussian as for the cubic. The only distributions for which the spread is very different are those 'with a sharp edge', i.e. those like the rectangular distribution for which the probability density falls abruptly to zero at a particular height. Distributions of this kind have not been found experimentally*. Thus we can say confidently that the highest asperity on a surface, or the separation at which contact first occurs, are poor reference points. Load-compliance curves based on the point of first contact may be expected to show scatter over their full range introduced by their origin and the more sensitive the apparatus so that contact is detected earlier, the more will be the scatter.

CONCLUSION

Any model of contact between surfaces, both of which are assumed to be rough, can be simulated by a model in which only one surface is rough.

Models with both surfaces rough, which take into account experimental findings on surface topography, give results indistinguishable from single-rough-surface models.

Nothing can be deduced from the load-compliance curve about the asperity shape or mode of deformation; the one clear prediction which can be made is that the scale of the separation is determined by the surface roughness, so that a load increase by a factor of about 50 will reduce the separation by one standard deviation of the joint roughness, or by $1.4 \times$ the standard deviation of the individual roughness in the case of two similar random surfaces.

* Even if they did exist, combining them to form the joint height distribution will eliminate the 'edge' so that the point of first contact will be subject to scatter.

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APPENDIX 1

THE LIMITATIONS OF ASPERITY MODELS

Equation (9) states that the real area of contact is proportional to $F_2(d/\sigma)$. When the load is increased sufficiently, the mean planes will overlap and d becomes negative. We can show that $F_2x \sim x^2$ when x is large and negative, suggesting that the real area of contact may be larger than the apparent area! This, of course, reflects a basic failure of asperity models: the cross-sectional area of an individual asperity becomes indefinitely large as the distance from its summit increases, because it is difficult to introduce mathematically the condition that the asperity must be bounded by its neighbours.

In a similar way, the number of contacts becomes infinite as $d \rightarrow \infty$ when in fact it should not exceed $(\eta\mathcal{A})^2$ and should begin to decrease as distinct contacts merge. The infinite value occurs because the model used implies contacts with asperities whose summits lie outside the area $\mathcal{A}$ considered.

These problems are unimportant provided only positive separations are considered. It is only the highest parts of a surface which can be modelled by isolated asperities of simple shapes; lower down the individual hills begin to link up just as mountains do on the surface of the earth.

The profilometric approach (17), despite its limitations, seems to offer the best way around this problem.

APPENDIX 2

INTEGRALS OF THE GAUSSIAN DISTRIBUTION

$$F_n(h) = \frac{1}{\sqrt{(2\pi)}} \int_h^\infty (u-h)^n e^{-u^2/2} du$$

These were calculated as described in (9). Useful properties are: for h large and positive

$$F_n(h) \sim \frac{n!}{\sqrt{(2\pi)}} h^{-n} e^{-h^2/2} \left\{ 1 - \frac{(n+1)(n+2)}{2h^2} + \dots \right\}$$

while for h large and negative

$$F_n(h) \sim (-h)^n$$

The recurrence relation $F_{n+1}(h) = nF_n(h) - hF_n(h)$ can be used to build up the higher functions.

APPENDIX 3

EXTREME VALUE DISTRIBUTIONS

Let p be the probability of an individual asperity height exceeding a particular value z . Then the probability that in a sample with n members all will lie below z is $(1-p)^n$.

A probability distribution is usually described by its mean and standard deviation, but it may equally be described by its median and inter-quartile range, where the quartiles and the median are the values such that $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{3}{4}$ of the members of the population lie below them. Unlike the mean and standard deviation, they are readily found for extreme value distributions: in fact we have simply to solve the equations

$$(1-p(z))^n = \lambda \quad \text{where } \lambda = 0.25, 0.50, \text{ or } 0.75$$

We are usually interested in the case of p small and n large, in which case $(1-p)^n \sim e^{-np}$ and

$$p(z) \sim -\frac{1}{n} \ln \lambda$$

For example, the distribution of the highest of a sample with 10^4 members will have a median and quartiles given by

$$p_{\text{lower quartile}} = -10^{-4} \ln(0.25) = 1.386 \times 10^{-4}$$

$$p_{\text{median}} = 0.6931 \times 10^{-4}; \quad p_{\text{upper quartile}} = 0.2877 \times 10^{-4}$$

To complete the analysis we have to specify the parent distribution from which the sample is drawn. For example, for the cubic height distribution used in the text, we have

$$p(z) = \int_z^{2L} \phi_0(z) dz = \int_z^{2L} \frac{2}{3L^4} (2L-z)^3 dz = \frac{1}{6} \left(\frac{2L-z}{L} \right)^4$$

for heights in the range $L < z < 2L$. If we measure heights from the mean $\mu = 2L/3$ and change from the range L to the standard deviation $\sigma = L/3$ this becomes

$$p = \frac{1}{486} \left(4 - \frac{z'}{\sigma} \right)^4 \quad \sigma < z' < 4\sigma$$

Thus, for the sample of 10^4 asperities, the median is where

$$\frac{1}{486} \left(4 - \frac{z'}{\sigma} \right)^4 = 0.6931 \times 10^{-4}$$

$$\text{i.e. } \frac{z'}{\sigma} = 4 - (486 \times 0.6931 \times 10^{-4})^{1/4} = 3.5716$$

and the quartiles are 3.4906σ and 3.6561σ .

h	$F_0(h)$	$F_{1/2}(h)$	$F_1(h)$	$F_{3/2}(h)$	$F_2(h)$	$F_{5/2}(h)$
0.0	0.500 00	0.411 09	0.398 94	0.430 02	0.500 00	0.616 64
0.5	0.308 54	0.225 34	0.197 80	0.195 20	0.209 64	0.240 40
1.0	0.158 65	0.104 15	0.083 32	0.075 67	0.075 34	0.080 56
1.5	0.066 81	0.039 88	0.029 31	0.024 64	0.022 85	0.022 86
2.0	0.022 75	0.012 48	0.008 49	0.006 65	0.005 77	0.005 42
2.5	0.006 21	0.003 16	0.002 00	0.001 47	0.001 20	0.001 06
3.0	0.001 35	0.000 64	0.000 38	0.000 26	0.000 20	0.000 17
3.5	0.000 23	0.000 10	0.000 06	0.000 04	0.000 03	0.000 02
4.0	0.000 03	0.000 01	0.000 01	0.000 00	0.000 00	0.000 00

APPENDIX 4

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